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Algorithmic Influences: Analyzing Variations in Youth Mental Health Effects Across Social Media Platforms During & Post-Covid-19

Shubhi Tiwari, Shivam Atulya

UG Students, BCA, Department of Computer Science, Jain Deemed-to-be-University, Bangalore, India

ABSTRACT*:* This study explores the impact of social media engagement on youth mental health during and after the COVID-19 pandemic, focusing on Instagram, TikTok, and Snapchat. Using a mixed-methods approach, it analyses survey data from 104 participants alongside existing literature. Findings indicate heightened anxiety and depression during the pandemic, with lingering effects post-pandemic. Platform-specific features, such as Instagram's peer comparison, TikTok's rapid content exposure, and misinformation, contributed to stress and anxiety. Algorithm-driven content curation also influenced mental health outcomes. The study highlights the need for ongoing monitoring and interventions to mitigate social media's negative effects while fostering healthier online behaviours.

KEYWORDS: Social media usage; Mental health impact; Youth behavior; COVID-19 effects; Anxiety and depression; Social media algorithms; Regression Analysis; Statistical Analysis

I. INTRODUCTION

The rapid growth of social media platforms has fundamentally transformed the way youth communicate, express themselves, and perceive their identities. This shift raises important questions about the impact of these platforms on mental health, particularly among adolescents who are deeply immersed in digital environments. Despite theincreasing interest in understanding the relationship between social media use and mental health, significant gaps remain in the literature. Many current studies focus on narrow age ranges or specific platforms, limiting the generalizability of their findings. Additionally, the influence of factors like social comparison, online behavior, and content engagement on mental health outcomes such as anxiety, depression, and body dissatisfaction is not yet fully understood. As young people navigate the complexities of their online identities, it becomes crucial to explore how social media shapes their self- perception and mental health across diverse demographics and cultural contexts. Understanding these dynamics is essential to developing effective interventions that promote healthier social media use and mitigate potential negative effects.

This research adopts a mixed-methods approach, integrating both quantitative and qualitative analyses to provide a comprehensive understanding of how social media influences youth experiences. Surveys and self-reported data were collected from a diverse sample of youth to assess their social media use, mental health symptoms, and self-regulation strategies. In addition, content analysis was conducted to examine themes prevalent in user-generated content on popular platforms, offering insights into the narratives and identities constructed online. Statistical techniques, including regression and correlation analysis, were employed to identify patterns in the data.

II. LITERATURE REVIEW

Recent studies have explored social media's impact on adolescent mental health, yet gaps remain. **Etkisi** [1] examined Instagram's role in self-representation but lacked cross-platform analysis. **Zheluk et al**. [2] linked TikTok content to anxiety but did not explore underlying psychological mechanisms. **Gómez Restrepo et al**. [3] highlighted mental health challenges during COVID-19 in a specific region, limiting generalizability. **Talukdar** [4] found significant links between social media use and mental health but lacked long-term data. **Dreier et al**. [5] studied mood regulation via

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daily surveys but had a narrow demographic scope. **Wiedemann et al**. [6] reviewed resilience factors but focused on early pandemic phases. **Lin** [7] assessed TikTok's psychological effects but ignored algorithmic influence. **Vanherle et al**. [8] linked Snapchat use to loneliness and depression, though its sample was predominantly female. **Beyari** [9] identified harmful social media features without comparing platforms. **Srivastava et al**. [10] used machine learning to study addiction but overlooked demographic differences. Most studies focus on single platforms or specific demographics, limiting their broader applicability. This paper addresses these gaps by analyzing platform-specific features and their mental health effects across diverse populations using a mixed-methods approach.

Year	Title & Author(s)	Approach	Objective	Key Findings	Limitations
2023	Instagram's Influence on Self-Representation of Young Users (Kendini Temsiline Etkisi)	Mixed-methods	Investigate Instagram's impact on self-representation	Affects self-perception	No long-term, cross-platform comparison
Dec-2022	Adolescent Anxiety and TikTok (A. Zheluk, J. Anderson, S. Dineen-Griffin)	Content Analysis	Explore TikTok's link to anxiety	Content influences anxiety	Lacks algorithm analysis
2023	Mental Health & Resilience in Adolescents During COVID-19 in Colombia (Carlos Gómez Restrepo et al.)	Statistical	Document mental health & resilience	Need for targeted support	Limited to specific populations
June 2023	Impact of Social Media on Adolescent Mental Health in Thailand (Jeuti Talukdar)	Survey (SPSS)	Investigate social media's impact	Linked to mental health	No long-term effect analysis
July 2024	Adolescent's Self-Regulation of Social Media Use (Melissa J. Dreier et al.)	Daily Surveys	Explore self-regulation & mood	Individual differences noted	Limited demographic diversity
2023	Pre-Pandemic Resilience Factors (A. Wiedemann, R. Gupta, C. Okey, Juli)	Systematic Review	Examine resilience during the pandemic	Family-level factors crucial	Limited early-pandemic focus
2023	Psychological Impact of TikTok (Zheng Lin)	Mixed-methods	Explore TikTok's impact on mental health	Shows addiction, cyber bullying	Needs further algorithm study
2023	Social Surveillance on Snapchat (R. Vanherle et al.)	SEM	Examine Snapchat's effect on loneliness	Links to loneliness, FOMO	Lacks gender diversity, no causality
2023	Social Media & Mental Health Problems (Hasan Beyari)	Analytical Process	Identify harmful social media features	Features ranked	No platform-specific comparisons
Not Defined	Social Media Addiction & Stress in Indian Youth (Dr. A. Srivastava et al.)	Machine Learning	Study addiction's impact on stress	Data-driven stress prediction	Lacks detailed algorithm analysis

Table 1: Summary Table Of Literature Review

III.METHODOLOGY

III.1 Data Collection Methods

This study employed a mixed-methods approach, integrating qualitative and quantitative data through an online survey and literature review to assess social media's impact on youth mental health during and after COVID-19.

III.1.1 Online Survey: An online survey was designed for youth aged 15–25, a demographic highly engaged with social media, especially during the pandemic. Hosted on Google Forms, it ensured accessibility and encouraged higher

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participation.

A Google Forms survey targeted youth aged 15–25, collecting data on:

- **Demographics:** Age, gender, and geographical location.
- **Social Media Engagement:** Platforms used, daily screen time, and interaction types (messaging, posting, content viewing).
- **Self-Reported Mental Health:** Anxiety, depression, and body image ratings on a 1–10 scale.

With 104 valid responses, descriptive statistics and correlation analysis identified trends and relationships.

III.1.2 Data Collection from Existing Literature

Peer-reviewed studies were analyzed to supplement survey findings, focusing on:

1. **User Engagement Behaviors:** Age, gender, platform usage, and interaction types.
2. **Mental Health Indicators:** Anxiety, depression, stress, self-esteem, and well-being.

User ID	Platform	Time Spent (hrs/day)	Content Type	Anxiety (1-5)	Depression (1-5)	Self-Esteem (1-5)	Likes	Comments
1	Instagram	5	Images	4	3	2	150	45
2	TikTok	2	Videos	2	4	1	200	60
3	Snapchat	3	Stories	5	2	4	120	30
4	Instagram	4	Stories	4	3	4	100	25
5	TikTok	1	Videos	3	4	2	250	70
6	Snapchat	2	Images	3	2	3	180	50
7	Instagram	3.5	Videos	5	3	2	300	80
8	TikTok	1.5	Videos	5	5	1	110	28
9	Snapchat	2.5	Stories	3	3	3	170	40
10	Instagram	3	Images	5	3	3	200	60

Table 2: Summary of User Demographics and Mental Health Indicators

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ID	Age	Gender	Platform	Time Spent (hrs/day)	Frequency of Use (sessions/day)
1	25	Male	Instagram	2.5	4
2	21	Female	TikTok	2.1	5
3	20	Female	Snapchat	1.8	3
4	18	Male	Instagram	2.2	4
5	16	Female	TikTok	2.8	6
6	17	Male	Snapchat	1.5	2
7	24	Female	Instagram	3.2	5
8	21	Male	TikTok	2.9	4
9	19	Female	Snapchat	2.1	2
10	18	Male	Instagram	2.7	4

Engagement Behaviors	Platform Features	Well-being	Stress	Anxiety	Depression	Body Image	Self-esteem
Likes, Comments	Stories, Filters	7	5	6	4	7	6
Videos, Duets	Filters, Feeds	6	8	5	3	8	7
Messages, Streaks	Filters, Stories	8	4	7	6	5	6
Likes, Comments	Stories, Filters	6	7	5	5	6	7
Shares	Feeds	5	6	5	6	7	6
Streaks	Stories	7	5	7	5	8	7
Comments	Filters, Feeds	5	7	6	5	7	6
Shares, Comments	Stories, Filters	8	4	5	7	6	5
Messages	Filters	6	5	7	5	7	7
Likes	Stories, Filters	6	5	7	5	5	6

Table 3: Comprehensive Data on Social Media Use

Tables summarize key insights, showing platform-specific emotional impacts, such as increased anxiety from high TikTok usage and improved well-being from educational content.

III.2 Data Analysis Methods

Data analysis was conducted using R programming language, employing:

- **Descriptive Statistics:** Summarizing engagement patterns and mental health outcomes using mean, median, and standard deviation.
- **Regression Analysis:** Examining social media features (e.g., content type, social comparison) as predictors of anxiety and depression.
- **Correlation Analysis:** Pearson's coefficient identified significant links, such as a positive correlation ($r = 0.6$) between filter usage and anxiety.

This integrated approach provides a comprehensive understanding of social media's psychological effects, informing future research and interventions.

IV. RESULTS AND DISCUSSION

IV.1 Identifying Platform-Specific Features: The analysis examined correlations between social media features and mental health outcomes, revealing key insights into their impact on anxiety and depression

1. Correlation with Anxiety:

- **Filters:** showed the highest positive correlation with anxiety ($r = 0.6$), suggesting heightened self-image concerns.

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- **Messaging Usage:** had a moderate correlation with anxiety (r ≈ 0.4), potentially linked to response pressure and misunderstandings.
- **Posting Frequency:** also correlated moderately with anxiety (r ≈ 0.4), possibly due to stress over online persona maintenance.
- **Peer Comparison:** showed a weak negative correlation (r ≈ -0.2), indicating it may provide validation rather than increase anxiety.

2. Correlation with Depression:

- **Peer Comparison:** had the strongest positive correlation with depression (r ≈ 0.5), suggesting frequent comparisons lead to low self-esteem and depressive symptoms.
- **Algorithmic Feeds:** correlated with depression (r ≈ 0.4), as curated content may foster feelings of isolation.
- **Messaging Usage:** showed a moderate correlation (r ≈ 0.4), indicating that while it enables social connections, it may also contribute to loneliness.

Social media features impact mental health differently—filters and posting frequency heighten anxiety, while peer comparison and algorithmic feeds are strongly linked to depression.

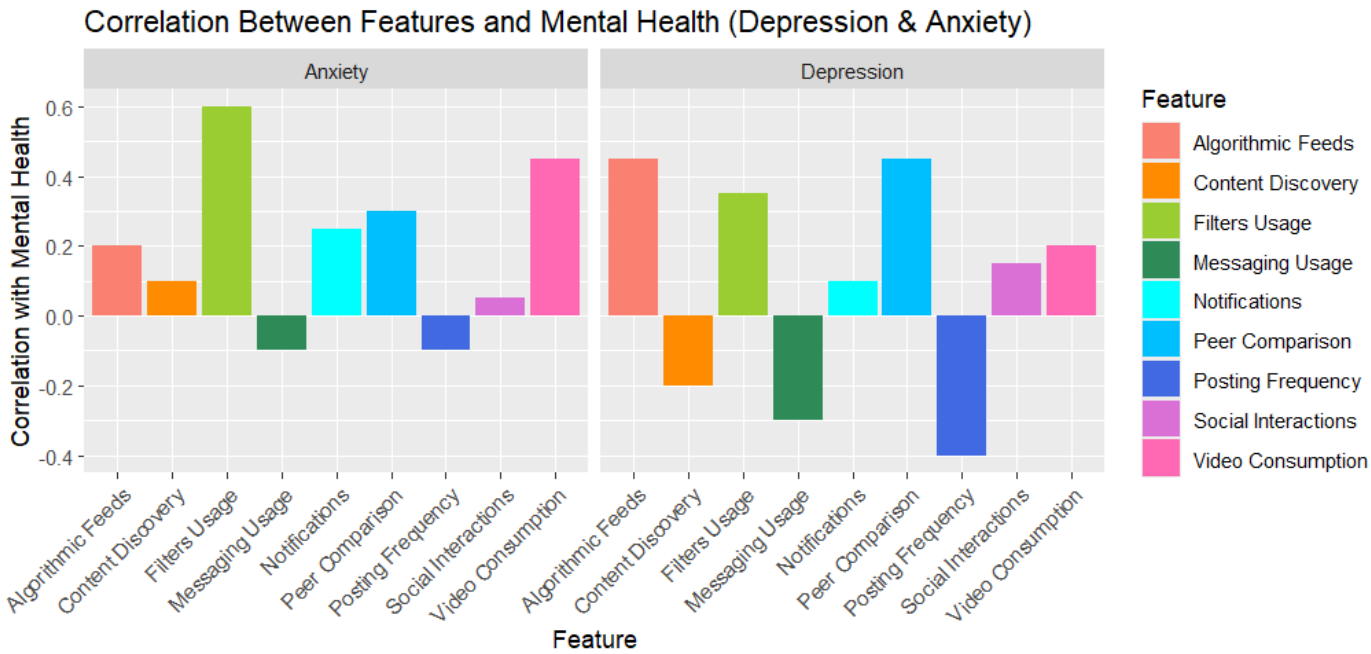

Fig 1: Correlation between Features and Mental Health

IV.2 Algorithms and Their Impact on Mental Health

This analysis examined the impact of content-based, ranking, and reinforcement algorithms on mental health across Instagram, Snapchat, and TikTok, identifying correlations with stress, anxiety, and depression.

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1. Types of Algorithms:

- **Content-Based Algorithms**: Personalize feeds based on user preferences but may increase exposure to distressing content.
- **Ranking Algorithms:** Prioritize popular content, influencing visibility and fostering competition.
- **Reinforcement Algorithms:** Adapt content based on user behaviour, potentially reinforcing negative engagement patterns.

2. Findings by Platform:

- **Instagram**: Ranking algorithms drive high stress due to visibility pressure, especially for content creators.
- **Snapchat:** Reinforcement algorithms contribute to anxiety as users strive to maintain engagement.
- **Content-Based Algorithms**: On both platforms, these algorithms heighten anxiety and depression by reinforcing unrealistic portrayals and echo chambers.

Personalized content enhances user experience but can also expose users to distressing material, creating echo chambers that reinforce negative self-perceptions.

3. Implications of Algorithmic Design:

This analysis highlights the significant impact of algorithmic design on mental health. Instagram's ranking algorithms and content-based algorithms on both Snapchat and Instagram correlate with increased anxiety and depression. These findings emphasize the need for platforms to integrate mental health considerations into their algorithms to foster healthier user experiences.

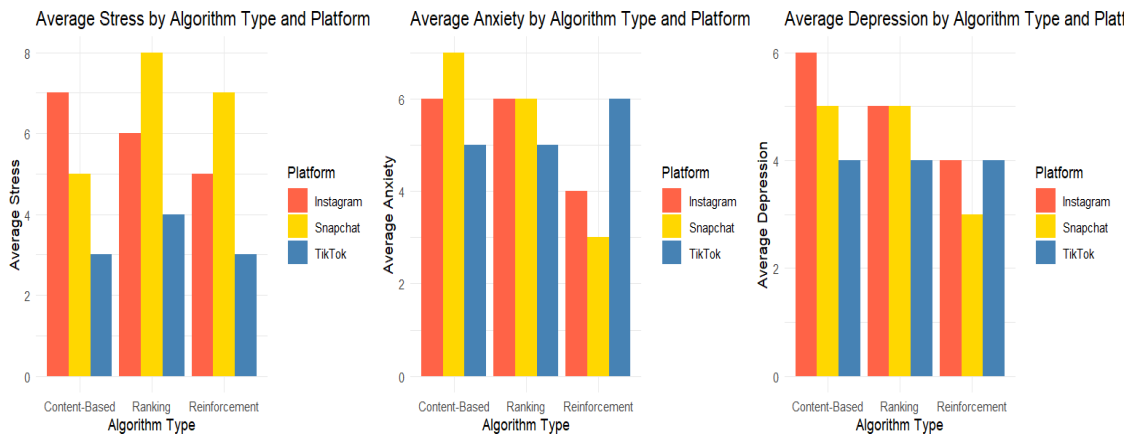

Fig 2 : Average Stress, Anxiety and Depression by Algorithm Type and Platform

IV.3 Usage Patterns During and After the Pandemic

This analysis examined variations in anxiety and depression levels across pre-pandemic, pandemic, and post-pandemic periods to assess the COVID-19 pandemic's impact on youth mental health and its lasting effects.

1. Mental Health Levels Across Time Periods:

- **Pre-Pandemic**: Anxiety and depression levels were moderate (avg. ~4/10).
- **During Pandemic**: Significant rise observed (avg. >7/10).
- **Post-Pandemic**: Slight decline but remained elevated (avg. ~5.5/10), indicating lasting effects.

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2. Implications for Mental Health Support:

The analysis highlights the critical need for continued mental health support for youth, even as the pandemic has subsided. Key considerations include:

- **Targeted Interventions:** Mental health programs tailored for post-pandemic challenges.
- **Awareness & Education**: Reducing stigma and encouraging help-seeking.
- **Access to Resources**: Ensuring availability for recovery and resilience.

In summary, the comparative analysis of mental health levels across pre-pandemic, youth mental health worsened during the pandemic, with anxiety and depression remaining above pre-pandemic levels. Sustained support and ongoing research are essential to address long-term impacts.

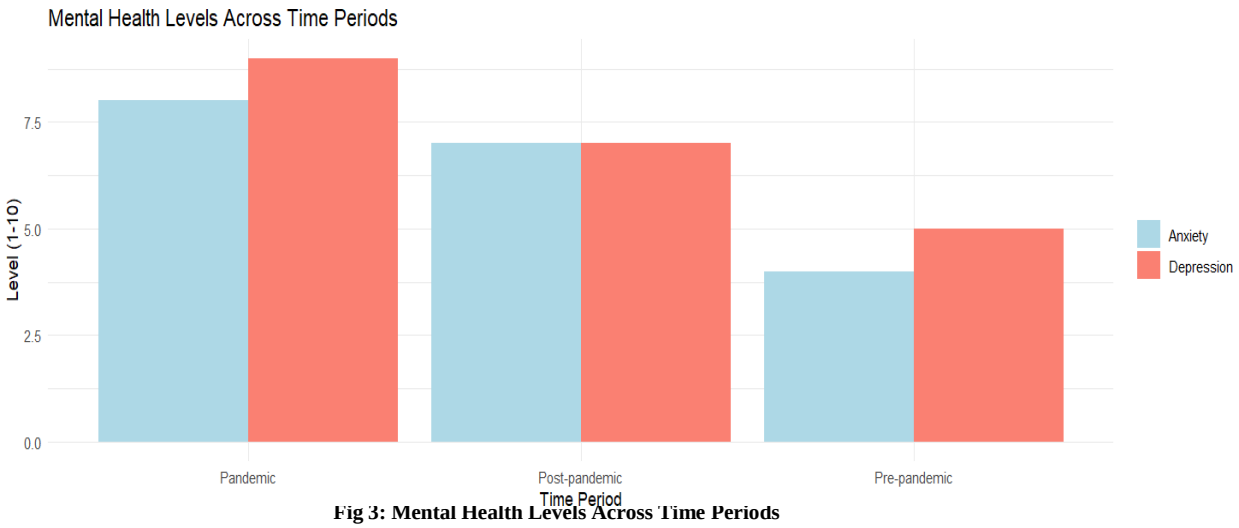

Fig 3: Mental Health Levels Across Time Periods

IV.4 Relationship Between Time Spent on Social Media and Anxiety

A scatter plot analysis using R's ggplot2 package examined the correlation between social media usage and anxiety levels, assessing whether increased usage correlates with higher anxiety.

1. Findings: The scatter plot displayed individual users' social media usage (x-axis) against their reported anxiety levels (y-axis), illustrating the distribution of data points and potential correlation.

- **Linear Regression Analysis**: The regression line showed a slight positive correlation, indicating a modest rise in anxiety with increased social media usage, though the relationship was weak.
- **Variability in Responses**: The analysis revealed significant variability, suggesting that factors like personal circumstances, social interactions, and content type influence anxiety levels beyond social media usage alone.

2. Implications of Findings: The findings suggest a modest association between social media use and anxiety, emphasizing the complexity of its impact. Individual differences, content type, and social interactions play crucial roles in shaping mental health outcomes.

- **Individual Differences**: Users' reactions to social media vary based on personality, coping mechanisms, and social context.
- **Need for Contextual Understanding**: Other factors must be considered when assessing social media's impact on mental health, rather than attributing anxiety solely to time spent online.

In conclusion, the analysis shows a slight positive correlation between social media use and anxiety, but the modest slope and variability highlight the need for a nuanced interpretation.

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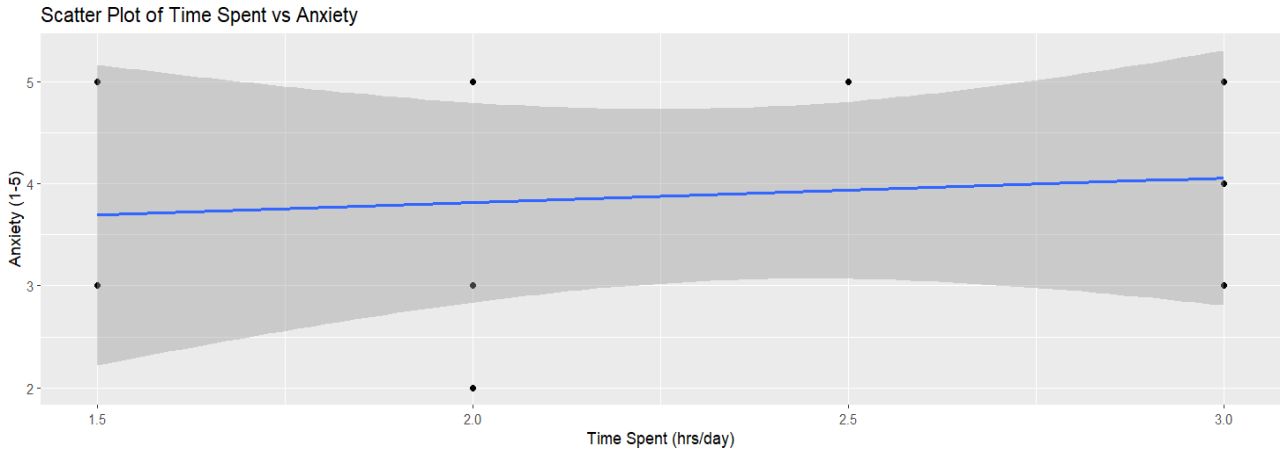

Fig 4 : Time Spent VS Anxiety

IV.5 Average Time Spent on Social Media

The analysis revealed varying daily usage patterns across platforms, with Instagram showing the highest engagement, highlighting potential mental health implications.

1. Usage Patterns by Platform:

- **Instagram**: Instagram had the highest engagement, with users averaging **4 hours per day**. Its visually curated content and influencer-driven culture may contribute to anxiety and body image concerns.**Snapchat**: Snapchat users averaged **3 hours per day**. Its ephemeral content fosters spontaneity but may also create anxiety over maintaining an image and responding quickly.
- **TikTok**: TikTok users averaged **2 hours per day**. While its short-form videos encourage creativity, the rapid content consumption may foster social comparison, impacting self-esteem and anxiety.

2. Implications for Mental Health: Differences in engagement levels indicate that platform design and content influence user experiences and mental well-being:

- **Instagram's Impact**: High engagement and emphasis on curated aesthetics may impact self-worth and body image, increasing anxiety and negative self-assessment.
- **Snapchat:** Ephemeral content fosters interaction but may heighten anxiety due to pressure to maintain engagement and peer validation.
- **TikTok:** Viral challenges encourage creativity but can lead to unhealthy comparisons, contributing to anxiety and self-doubt.

In summary, Instagram leads in engagement at 4 hours daily, with platform-specific features shaping user experiences and impacting mental health.

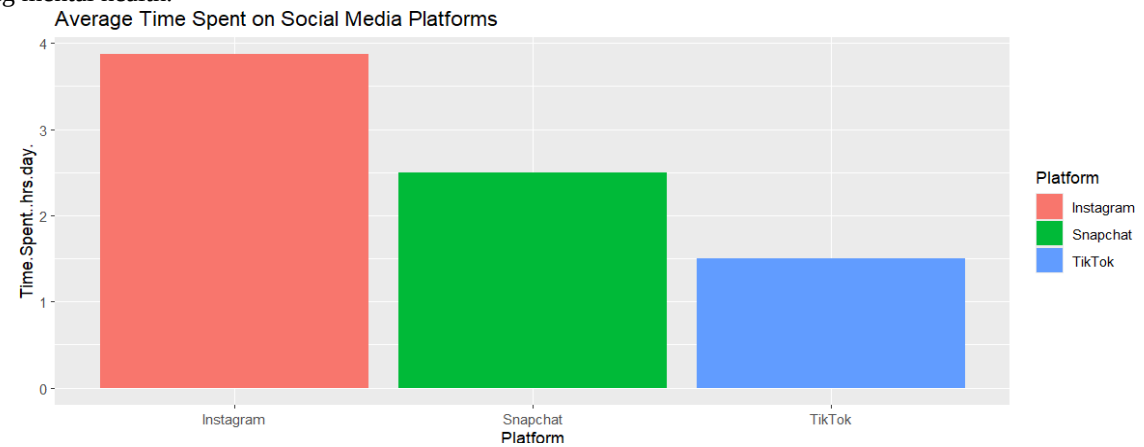

Fig 5: Average Time Spent On Social Media Platform

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V. OUTCOMES

The study revealed that different social media platforms impact youth mental health in unique ways, with Instagram fostering anxiety through peer comparison, TikTok affecting attention and stress levels, and Snapchat influencing self-esteem. Algorithm-driven content personalization creates echo chambers that reinforce negative mental health effects, while ranking algorithms increase stress due to visibility pressure. Features like filters contribute to body image issues, notifications drive compulsive usage, and engagement metrics affect self-worth. These findings highlight the need for platform-specific interventions to mitigate adverse mental health impacts on youth.

VI. CONCLUSION AND FUTURE WORK

This study examines the impact of social media on youth mental health during and after COVID-19, linking increased use of platforms like Instagram and TikTok to higher levels of anxiety and depression. Features like peer comparison, idealized self-presentation, and TikTok's rapid content can lead to negative self-perception and heightened stress. The spread of misinformation during the pandemic has further deepened confusion and mental health challenges among adolescents. The findings stress the need for mental health interventions focused on digital literacy and resilience to help youth manage social media's influences effectively. The study also highlights gaps in understanding the long-term effects of social media across diverse demographics, suggesting that future research take a longitudinal approach to explore these impacts. As social media remains central to youth lives, fostering a healthier digital space is crucial for their well-being.

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